Topic for Master 1/Master 2 internship

Academic year 2026-2027

Solving the S-matrix with machine learning: how fast can quantum interactions grow?

Scientific description:

Context

At weak coupling, quantum field theory has given us a remarkably complete picture of the consistent theories of interacting particles: they are built on gauge theories, and together they make up the Standard Model. At strong coupling, we do not know. What is the space of consistent quantum theories there, a space that also contains string theory, our best-understood consistent completion of gravity, and how can we compute anything in it? This is one of the central tasks of 21st-century theoretical physics.

Why is it so hard? The S-matrix, which encodes the outcome of every possible collision between particles, must conserve probability. This condition, unitarity, is a completely nonlinear equation (schematically $S\,S^\dagger = 1$) that ties all processes together. At weak coupling it can be solved order by order; at strong coupling we have almost no tools. For several years, the supervisor has been developing a new approach to this problem, participating in the revival of the S-matrix bootstrap of the 1960s, which gave birth to string theory, with modern machine-learning optimisation. We have constructed, for the first time, explicit four-dimensional amplitudes that conserve probability exactly in two-particle collisions while respecting relativistic invariance and causality, and that include the production of new particles through a tunable input [1–4].

This opens many avenues. One stands out as perhaps the most basic question one can ask about an interaction: how fast can it grow with energy? Since the 1960s we have known a universal speed limit, the Froissart bound: in theories with a mass gap, the total cross section cannot grow faster than the square of the logarithm of the energy. Yet we still do not know how genuine interacting theories approach this limit. Nature seems to come close: protons look bigger the harder they are smashed, all the way to LHC energies, a growth attributed to a mysterious collective exchange called the Pomeron. String theory, by contrast, is radically different at high energy, with collisions that become exponentially soft, and historically the graviton was first discovered hiding inside the Pomeron of the early string models.

Subject

The goal of the internship is to use these new tools to explore how quantum interactions behave at very high energy. After learning the basic principles of the S-matrix approach, the student can go in two directions, analytic or numerical, or combine both, depending on their interests. On the analytic side, the student will study how the nonlinear unitarity equation generates growth at high energy, in simplified models where the mechanism can be followed by hand. On the numerical side, the student will use our machine-learning solver to construct full amplitudes and measure how fast they grow.

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The core of the project is to understand how particle production controls the high-energy growth of amplitudes. How fast can they grow? Is the Froissart bound approached, and how? Do Regge behaviour (the collective exchange of whole towers of particles of increasing spin) and something resembling the Pomeron emerge? And how do these amplitudes compare with those of string theory? Answering these questions would shed new light on the space of consistent quantum theories and on the place of string theory within it.

A good background in quantum field theory is expected; an interest in numerical methods is welcome. Programming experience (e.g. Python) is useful but not required, and no prior experience with machine learning is needed.

Team and environment

The student will join a small, friendly team of postdocs and PhD students working on these questions, sharing their office, with regular one-to-one supervision and weekly group meetings where questions of any level are welcome. The group runs a journal club and regular informal discussions, and attends the lab seminars as well as the "Neural Collisions" seminar series on physics and machine learning. The student will also have opportunities to visit CERN, one hour away, where part of this work is done in collaboration. LAPTh is located in Annecy, between the lake and the mountains, and student housing is typically available for interns.

Opening towards a PhD thesis

The internship naturally opens to a PhD on the non-perturbative S-matrix. The thesis would extend the method to massless particles and ultimately to gravity, starting in more than four dimensions where gravitational scattering is better behaved, to understand how quantum gravity behaves in collisions far beyond the Planck scale and whether string theory emerges from first principles. Applications to the strong interactions, such as pion scattering, offer a complementary direction where results can be confronted with experimental data.

References

[1] P. Tourkine, A. Zhiboedov, *Scattering from production in 2d*, JHEP 07 (2021) 228 [arXiv:2101.05211]

[2] P. Tourkine, A. Zhiboedov, *Non-perturbative S-matrices from dispersive iterations*, JHEP 11 (2023) 005 [arXiv:2303.08839]

[3] M. Gumus, D. Leflot, P. Tourkine, A. Zhiboedov, *The S-matrix bootstrap with neural optimizers I: zero double discontinuity* [arXiv:2412.09610]

[4] M. Gumus, D. Leflot, P. Tourkine, A. Zhiboedov, *Neural S-matrix bootstrap II: solvable 4d amplitudes with particle production* [arXiv:2601.22145]

Possible extension as a PhD: YES

Keywords: S-matrix bootstrap, machine learning, high-energy scattering, Regge theory, Pomeron, Froissart bound, string theory, quantum gravity

Supervisor(s):

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